

Lab-4 Ex6 - Phone Number Analysis

Let's create a phone number analyzer! Complete the given program to perform the following:

1. Ask user to input a phone number. In Hong Kong, phone numbers are normally 8 digits, and the first digit cannot be 0 or 1. Assume the user always input an integer (*a proper C int value, but can be negative, fewer or more than 8 digits*).
2. Validate the phone number. Reject negative input, input with fewer or more than 8 digits, and input starting with 0 or 1. Print "Invalid phone number" and terminate the program in such cases.
3. Extract and store the occurrences of digits in the phone number.
4. Find the number of unique digits in the phone number.

Print out the results.

We have put the hints inside the pseudocode in `main.c`! You will also find that it is very similar to one of the example we have in this Week's lecture.

User input is ***bold and italic*** in the following sample runs.

Sample run #1:

```
Input phone number: 99999991
There are 2 unique digit(s).
```

Sample run #2:

```
Input phone number: 98765432
There are 8 unique digit(s).
```

Sample run #3:

```
Input phone number: 12345678
Invalid phone number
```